TAMASOMA JYOTIRGAMAYA
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VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

Professional Ethics and Human Values: Professionalism and Engineering in the Societal Context

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Unit 3

Syllabus

- Profession and Professionalism: Definitions; Professional Characters, Traits, Virtues, Roles, Rights and Responsibilities; Overview of Business Ethics and Work Ethics; STEM Professionals as Managers, Leaders, Consultants, Expert Witnesses, Scientists, Researchers and Academicians, and Entrepreneurs
- X` Profession as Social Experimentation: Comparison with scientific experiments; Application of Ethical Theories and Ethical Inquiries in professional contexts to gather relevant information; Informed Consent
- Responsibilities of STEM Professionals as Social Experimenters: Need for being Ethical by Design; Accountability and Answerability in general and in autonomous technology applications; Focus on DEIA, SDG, and Social Innovation; Guidance through Professional Societies

Understanding Professionalism

Professional Traits, Roles, Responsibilities, Virtues, Rights

Definitions

- Profession: Any job/occupation that requires advanced expertise (skill/knowledge), self-regulation, and concerted service to the public good.
- Professional (adj./n.): Of or pertaining to profession.
- Professionalism:
  - Attitude and typical qualities of a professional
  - Services related to achieving the public good, in addition to the practices of the knowledge of moral ideals. – Alasdair C. MacIntyre (Scottish Philosopher)

Naagarazan, R.S., "A Textbook on Professional Ethics and Human Values", New Age International Ltd., 2006

Being Professional

- Advanced Expertise
  - Sophisticated skills and theoretical knowledge for objective analysis of problems in known area
- Self-regulation
  - Independent of self-interest and towards the best interest of client/customer/end-user, sometimes guided by Codes of Conduct of the professional bodies
- Public Good
  - Not merely a salaried employee doing what the employer dictates. Working towards better welfare, safety, and health of society

Characteristics of Professional

- Proficiency
  - Extensive Training
  - Knowledge and Skills
  - Monopoly
    - Through degree, accreditation, and/or licence
- Self-regulation
  - Autonomy in workplace
- Public Welfare
  - Requires Ethical Standards

Roles & Responsibilities of a Professional

Roles

- Saviour
- Guardian
- Bureaucratic Servant
- Social Servant
- Social Enabler (Catalyst)
- Game Player

Responsibilities

- Moral Responsibility
  - Obligations
  - Conscientiousness
  - Liability and Accountability
  - Praiseworthiness/ Blameworthiness
- Causal Responsibility
- Job Responsibility
- Legal Responsibility

Professional Virtues

- **Proficiency**
  - Mastering technical skills (“intellectual virtue”), competence (qualification licence, commitment), diligence, creativity, excellence, self-renewal
- **Self-direction/Self-governance**
  - Understanding & Cognition: Self-understanding, humility, good moral judgement (“practical wisdom”); Commitment & Action: courage, self-discipline, perseverance, self-respect, integrity; honesty
- **Public-spirited**
  - Respect for rights of others, non-malfeasance (not intentionally harming), beneficence (preventing/removing harm), public good, unbiased justice.

Professional Virtues (Contd.)

- Team-work
  - Working with others, collegiality, cooperativeness, communication, respect for legitimate authority, ability to motivate others, responsible exercise of authority
- Cardinal (Chief)
  - Wisdom: Perception of truth and discernibility of right and wrong
  - Courage: Firm and elevated mind to take right action
  - Temperance: Order/Moderation in words and action
  - Justice: Preserving humanity and observing faith of contracts

Professional Traits

- Collegiality
- Loyalty/ Trustworthiness
- Confidentiality
- Respect for Authority

Also…

- Sense of Responsibility
- Commitment
- Conscientiousness
- Self-improvement
- Preparedness
- Leadership and Managerial Skills

- Also reflected in:
  - Appearance
  - Attitude
    - Courtesy
    - Composure
    - Respect
    - Conflict Resolution
  - Communication
    - Oral & Written
    - Phone/E-mail Etiquette
    - Meetings

Subramanian, R., "Professional Ethics", Oxford University Press, 2013

Professional Rights

- Human Rights
- Moral Rights
- Intellectual Property Rights
- Right to express professional judgement
- Right to talk about one's work publicly within the bounds of confidentiality
- Right to be associated with professional bodies and be involved in their activities
- Right to protect clients and public from dangers of one’s work
- Right to professional recognition and due credit for one’s services
- Right of professional conscience to fulfil professional responsibilities
- Right to whistleblowing to preserve public safety and well-being

Other Rights of Employees

- Right to refuse involvement in illegal and unethical activities
- Right to criticise the organization (Right to free speech and dissent)
- Right to be involved in activities of choice outside the organization, as long as no conflict of interest arises
- Right to privacy
- Right to fair enquiry in case of any alleged violations of Codes
- Right for institutional recognition of rights

Ethics in Business and Workplace

Business Ethics

- Study of moral issues arising in exchange of goods and services that are fundamental to our existence
- The 3 Cs
  - **Compliance:** Laws ("Rules"), Morality, Company policy
  - **Contribution:** Core values, Product/Service quality, Employment, Spinoffs, Quality of Work Life (QWL)
  - **Consequences:** Towards environment, Public image, Social responsibility

Paliwal, M., "Business Ethics", New Age International, 2006

Engineering as a Profession

STEM Professionals in various roles

Motivation to study Engineering Ethics

- A successful engineer, in contributing to the character of the engineering profession, according to T.P. Hughes:
  - Is orderly, regular in one's habits, disciplined, predictable, methodical in problem-solving, even-tempered, and law-abiding
  - [Brings] order out of the chaos of [one's] natural instincts

Hughes, T.P., Introduction to "Selections from Lives of the Engineers" of Samuel Smiles, MIT Press, 1966

Work Ethic

- Sources of ethical guidance such as laws (“rules”), social conventions, professional codes of conduct and religious beliefs
- Identifying the ethical values integral to the professions
- Obligations these values give to members of the professions
- Ethical issues which arise when they are concerned to produce benefits and prevent harm
- Treat people fairly, respect others and act with integrity
- How these values might be incorporated into professional practice
- Dealing with blame in a professional context and claims about rights

Engineers and Ethical Issues

- Multiple roles in the same project
  - Engineer, Manager, Team Leader, Consultant
- Direct and Indirect impact on Society, Environment, Lifestyles, Interactions
- Assessment of Safety and Risk
- Nature of projects involving engineers
  - Developing Weapons of Mass Destruction
  - Privacy and Confidentiality
  - Cyber crimes (for Intelligence and Defence Departments)
  - Artificial Intelligence
  - IoT
  - …

http://www.iep.utm.edu/ap-ethic/#SH5b

Objectives of Ethics (Revisited)

- To study human behaviour and make an evaluative assessment about them as moral or immoral
  - To establish moral standards and norms of behaviour ("systematize")
  - To make judgment upon human behaviour based on these standards and norms ("defend")
  - To prescribe moral behavior and recommend dos and don'ts of behaviour ("recommend")
- Ethics help form an opinion or attitude about human conduct in general

Objectives of Engineering Ethics

- Understanding Moral Values of Engineers/ Engineering Organizations
- Resolving Moral Issues in Engineering profession/ organizations
  - Also includes legal, political, social, economic, cultural aspects
- Justifying Moral Judgements concerned with Engineering practice

- Micro Ethics: Ethics of engineers day to day in an organization
- Macro Ethics: Ethics of engineers in a regional/ national level

Engineering Profession

- Professionalism must extend all through
  - Conception
  - Design
  - Manufacture
  - Testing
  - Sales
  - Service/ Maintenance
- Is the product…
  - Safe and really useful?
  - Designed to last as long as it can?
  - Harmful in the long run?
  - Made with material justified for its quality vs. cost/budget?
- Is the engineer…
  - moral and ethical?
  - adhering to the professional Code of Ethics and all laws and policies?

Interactions of Engineers

- Engineers must interact with
  - Colleagues (Superiors/ Managers), Peers, Subordinates
  - Family members
  - Engineering Professional Bodies
  - Global Environment
  - Industry and other firms
  - Clients/ Customers (End-users)
  - Bodies of Law & Order and Government

STEM Professionals as Managers

STEM Professionals as Managers

- Engineers are "professional" (Ref. Professional Virtues, Roles, Responsibilities, Characters)
- Engineers can learn managerial skills easier than managers learn engineering skills
- Management requires knowledge of finance, accounting, scheduling, …
- Managers have responsibilities towards employees, customers/ end-users/ clients, dealers/ vendors/ suppliers, general public
  - "Professionals" have technical expertise and relevant decision-making skills, self-regulation to manage things for greater benefits (and not just for profits), and ensure public welfare, safety, and health
- Responsibilities: Promoting "Ethical Environment" and Conflict Management

Promoting Ethical Climate/ Environment

- Work environment conducive to good moral conduct & ethical behaviour
- Tata, Birla, Reliance, TVS, L&T, Google, …
- Himalaya had an Ethics Director as a Vice President, reporting to Ethics Committee!
  - Addresses employees' ethical concerns, assesses their degree of ethical awareness, conducts GDs and workshops on ethics, wrote weekly articles regarding specific cases and concerns, available to all employees on a confidential phone line – "Ethics Ombudsperson"
- Ethical Values must be appreciated by managers and employees alike
- Ethical language as a part of "corporate dialogue"
- Top Management as role model in both letter and spirit and personal examples
- Clear procedures for conflict resolution, say, through Ethical Ombudspersons

Raghavan, M.P., "Professional Ethics and Human Values", Scitech Publications India Pvt. Ltd., 2013

Moral Problems in Management

- Moral problems are truly managerial dilemmas: a conflict between an organization's economic performance (measured by revenues, costs and profits) and its social performance (stated in terms of obligations to persons both within and outside the organization).
- Most ethical decisions have:
  - extended consequences
  - multiple alternatives
  - mixed outcomes
  - uncertain consequences
  - personal implications

http://www.karlknapp.com/resources/ethics/ethicsofmgmt.html

Ethics of Management: Decision Checklist

- What are the best economic alternatives?
- What are the legal alternatives?
- Does a given decision result in greater benefits than damages for society as a whole, not just for our organization as part of that society?
- Is the decision self-serving, or would we be willing to have everyone else take the same action when faced with the same circumstances?
- We understand the need for social cooperation; will our decision increase or decrease the willingness of others to contribute?
- We recognize the importance of personal freedom; will our decision increase or decrease the liberty of others to act?
- Lastly, we know that the universe is large and infinite, while we are small and our lives are short; is our personal improvement that important, measured against the immensity of that other scale?

http://www.karlknapp.com/resources/ethics/ethicsofmgmt.html

Principles of Ethical Decision Making (Revisited)

- Principle of Long-term self-interest
  - Never take any action not in your organization's long-term self-interest
- Principle of Personal Virtue
  - Never do anything that is not honest, open, and truthful and that you would never be glad to see reported in the newspapers or on TV
- Principle of Religious injunction
  - Never take any action that is not kind and that does not build a sense of community.
- Principle of Government Requirements
  - Never take any action that violates the law, for the law represents the minimal moral standard
- Principle of Utilitarian benefit
  - Never take any action that does not result in greater good for society
- Principle of Individual rights
  - Never take any actions that infringes on others' agreed-upon rights
- Principle of Distributive Justice
  - Never take any action that harms the least among us: the poor, the uneducated, the unemployed

Paliwal, M., "Business Ethics", New Age International, 2006; http://amr.aom.org/content/20/2/379

STEM Professionals as Leaders

STEM Professionals as Leaders

- Moral (Morally Creative) Leadership
  - As academicians, managers, government servants, or bureaucratic servants
  - Contributions to communities, professional societies, and to the profession in general
  - Leading a group of people towards a common goal (involving innovation) without ignoring morals

Raghavan, M.P., "Professional Ethics and Human Values", Scitech Publications India Pvt. Ltd., 2013

Ethical Leadership

- Problems cannot simply be fixed going forward …ignoring the history!
- Ethical mistakes do not go away if ignored. They worsen the situation!
- Ethical climate is a must to understand ethics before owing/implementing
- Ethical leadership requires a proactive approach – you think for your organization!
- Ask: "Is it legal? Is it still unethical?" – Correct for ethics if necessary.
- Ethics must be kept independent and autonomous – Ref. Ethics Ombudsperson
- "Ethical leadership is not just about having and acting on sound values; it is about confronting the facts no matter how uncomfortable it may be to do so." – Mark Pastin, CEO-Council of Ethical Organizations

Mark Pastin, http://chiefexecutive.net/biggest-ethical-mistakes-made-ceos-avoid/

STEM Professionals as Consultants and Expert Witnesses

STEM Professionals as Consultants

- Freelancers – paid fees for service to clients and not salary by employers
- Areas of work where ethics are particularly important for Consulting Engineers
  - Advertising
    - Must not involve lies, half-truth, exaggeration, falsities
    - Must not affect mutual respect or respect towards profession
  - Competitive Bidding
    - Quoting price for project confidentiality is against Code of Ethics
    - Reduction is quality can occur due to lower bidding
  - Contingency fees
    - Paid if client's money is saved if a possible but unpredictable event is avoided by Consulting Engineer (Ref. Lehman's formula or 5-4-3-2-1 formula)
  - Resolution of disputes
    - Time-taking – can affect service quality ultimately
    - Avoidable – any definite necessity must be resolved in the best interests of the client

Raghavan, M.P., "Professional Ethics and Human Values", Scitech Publications India Pvt. Ltd., 2013

Further Reading

- Koontz, H.; O'Donnell, C., "Principles of Management: An Analysis of Managerial Functions", McGraw-Hill (1968)
- http://nraomtr.blogspot.in/search/label/Business%20and%20managerial%20ethics
- https://sol.du.ac.in/mod/book/view.php?id=1546&chapterid=1531

http://nraomtr.blogspot.in/2011/12/principles-o-principles-of-management.html

STEM Professionals as Expert Witnesses

- Sometimes required for expert witness in the Court of Law
  - Expert witness: One who can certify based on facts and figures in an area of expertise
    - Need not be an eye-witness but can be considered witness for analysing and inferring from available information/ data
    - Must maintain confidentiality of the case
    - Must maintain an unbiased stand as a witness/ expert
    - Must not be "hired guns" – giving false information by receiving consulting fees
    - Must not be biased by finance/ money, ego, or sympathy
    - Must not show loyalty to one's own clients
- Cases can be of any nature involving engineering expertise to arrive at a better understanding of the situation
  - "What caused the failure of …?", "What could be the cost involved in …?"

Raghavan, M.P., "Professional Ethics and Human Values", Scitech Publications India Pvt. Ltd., 2013

STEM Professionals in Science Research and Academia

Responsible Innovation and Accountability

Responsible Research and Innovation (RRI)

- *Responsible research and innovation is an approach that anticipates and assesses potential implications and societal expectations with regard to research and innovation, with the aim to foster the design of inclusive and sustainable research and innovation.*

https://ec.europa.eu/programmes/horizon2020/en/h2020-section/responsible-research-innovation

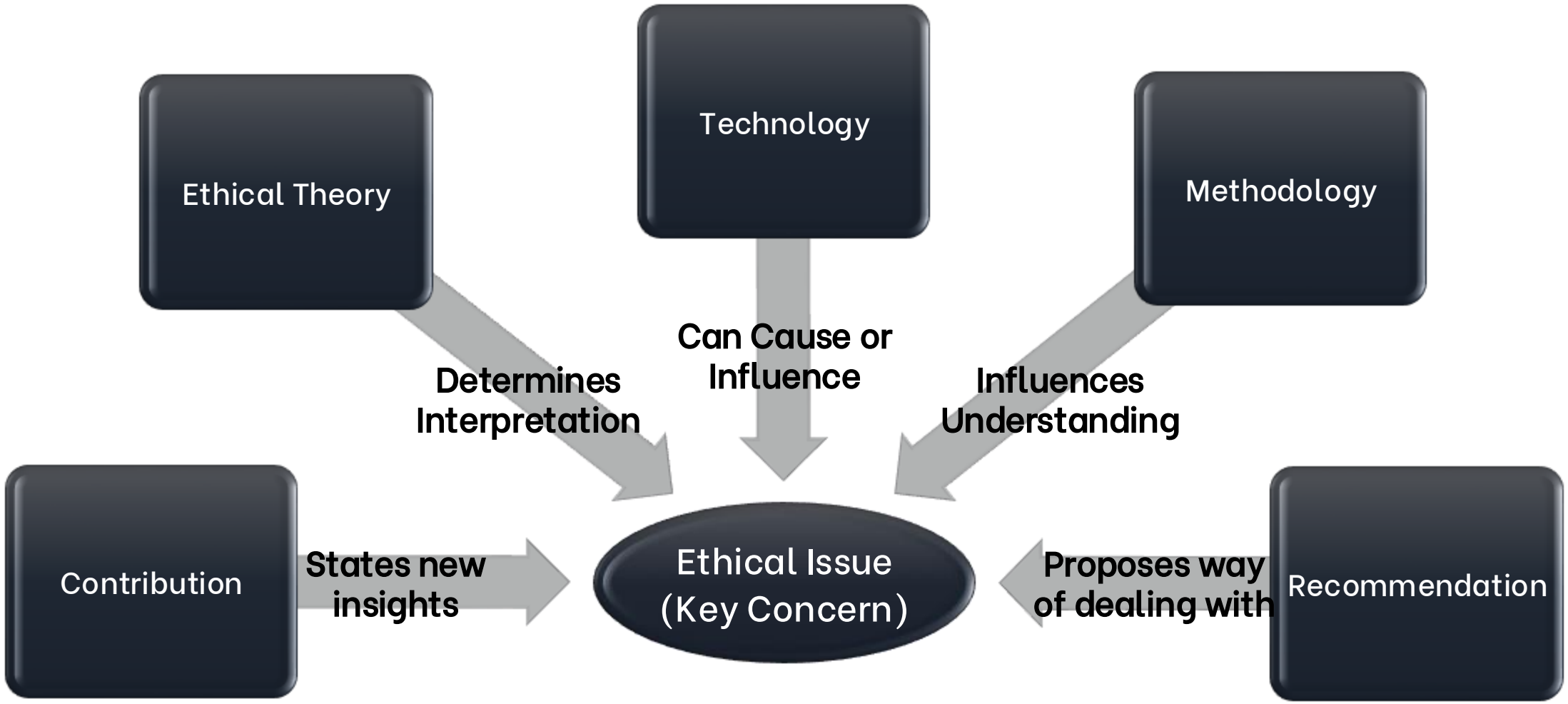
Tools to help resolve Ethical Issues

- https://www.dporganizer.com
  - Map, visualize, report and manage your processing of personal data – ensure General Data Protection Regulation (GDPR) compliance
- https://onetrust.com
  - Create PIA and data mapping to determine privacy risk and help incorporate privacy into design
- https://ethicscanvas.org
  - Structure ideas about ethical implications, visualise, resolve collaboratively
- https://privacyperfect.com
  - Ensure Data Protection Compliance

Bringing in More Accountability

- Establishing Trust in Public Consultation
  - Informed Consent, Ethical Guidelines for Research, ...
- Making Ethics a part of the architecture and not afterthought!
  - Ethical by design
  - "Security mindset": *Rather than thinking about how a system will work, imagine how it could fail.* — Bruce Schneier, Cryptographer
- New ontology to be explored
  - Being human first: Understanding beyond our current notions of democracy, policy-making, economics, law, privacy, safety, security, identity, etc.
  - Privacy Impact Assessment methodologies and validation metrics needed – very important for healthcare applications

Approaching Ethical Issues



Stahl, B.C.; Timmermans, J.; Mittelstadt, B.D., "The Ethics of Computing: A Survey of the Computer-Oriented Literature", *ACM Computing Surveys*, 48(4):55, Feb. 2016

Ethical by Design: Principles



- What might be the effects of our products being used at greater scale than we anticipate? How will this product change the world and people's lives?
- How will this technology affect people's lives and communities whether they are using the products or not? Will this encourage prosocial or antisocial behaviour?
- How might our design harm people? Will this technology replace human workers or relationships? How can we reduce the likelihood and scale of this harm?
- What is the worst possible way in which our technology could be used or repurposed?
- What would happen if our worst fears regarding this product come to pass?

- Harm Metrics
- Harm Quarantines

https://ethics.org.au/ethical-by-design/

STEM Professionals as Entrepreneurs

Human Values for Entrepreneurs

- Empathy
  - Helps understand pain points of users
  - Helps study market and identify scope
- Conscientiousness (అంతఃకరణశుద్ధి) → Integrity (త్రికరణశుద్ధి)
  - The intention to do the right thing
  - Leads to integrity – saying what you think and doing it
- Trust
  - Absolutely necessary for a brand to survive!

Being Empathetic

- Open-minded
- Observant
- Listening keen
- Curious
- Assuring

- Create trust
  - Why should anyone tell *you*?
- Look from the other side
  - Can you be on *that* side?
- Consider emotion
  - Are you similarly emotional?
- Do not judge
  - Can you …*just accept*?

Unethical Acts of Organisations/ Startups

- Pricing
- Unfair salary practices
- Creating artificial demand
- Use of proprietary or scarce materials
- Adding unethical costs to society
- Lack of transparency
- Incrementalism
- Self-serving bias
- Covering up mistakes

A To-Do List for Entrepreneurs

- Establish Ethical Climate, top-down, built on unambiguously-defined ethical framework
- Ensure that organisational policies and values do not require unethical behaviour
- Encourage and ensure open communication
- Provide forum for resolving ethical dilemmas, while protecting individual identity
- Create uniform framework for fulfilling moral responsibilities of punishing and rewarding bad and good organisational behaviour and performance
- Conduct ethics training periodically
- Conduct ethics audits and appraisals

Related Topics (Covered Elsewhere in Curriculum)

- Ethico-economic system
- Conflicts at workplace
- Ethics for Managers
- Ethics for Organisations

STEM Profession as Social Experimentation

Social Experiment

- Social:
  - relating to society or its organization
  - Involves people – as one or more communities or as individuals
- Experiment:
  - a scientific procedure undertaken to make a discovery, test a hypothesis, or demonstrate a known fact
- Social Experiment
  - a research project conducted with human subjects in the real world …

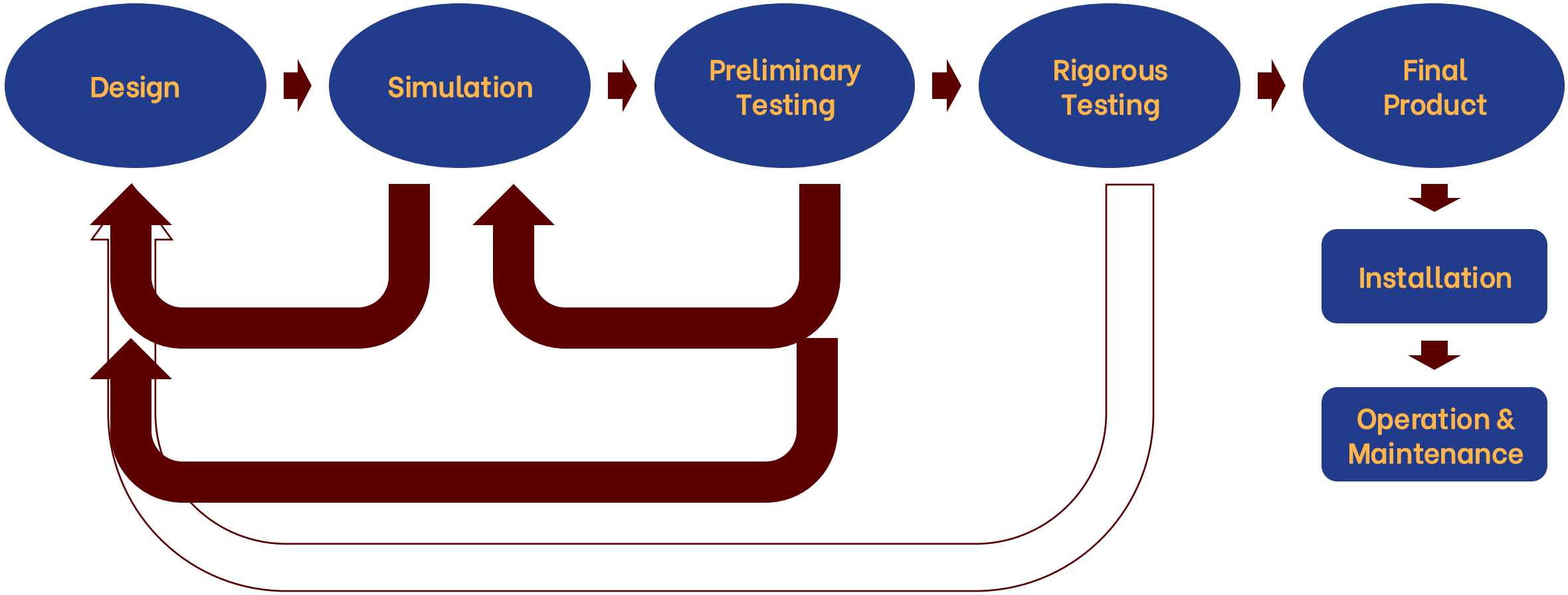
Engineering as Experimentation

- **Design:** Tests and simulations
  - Formal experimentation with various materials and methods – preliminary testing
  - Rigorous testing after the final design is chosen based on preliminary results
- **Production:** More tests before final product is created
  - Must ensure "compliance" and "consequences"
- **Installation**
- **Operation & Maintenance**

Aspects of Social Experimentation

- Society and its components are the "subjects" of experimentation
  - Experimentation because all outcomes and consequences, benefits and risks, *etc.* are not always known
- What the "experimenters" cannot know beforehand may result in unanticipated or uncertain outcomes
  - Risk involved due to lack of knowledge about outcomes
  - Decision-making may be blinded or biased by lack of knowledge
- People are the subjects of experiments
  - Rights of Subjects and Duties of Experimenters are both important to be considered from ethics point of view
  - Risk Assessment and Costs vs. Benefits are both important
  - Legal Obligations of Experimenters may limit innovation
  - Codes of Ethics put primary responsibilities on Experimenters

Engineering as Experimentation



Factors affecting "Experiments"

- Phase I: Design – Testing – Production
  - Time schedules
  - Monetary investments
  - Confidentiality
- Phase II: Installation – Operation – Maintenance
  - End-user Satisfaction
  - Return on Investments (RoI)
  - Unintended Outcomes
  - Uncertainties
  - Compliance procedures
  - Consequential scenarios: Social, economic, environmental impact

Possible Failures

- Incomplete understanding of implications
  - Insufficient time or money
  - Commercial advantages due to confidentiality
  - Uncertainty about impacts
- Participation of subjects
  - “Non-engineers” – No control group
  - Responsibility on subjects if risk accepted voluntarily
- Outcome Assessment
  - Improvements to the product
  - Risk management

Subjects & Impacts

- Subjects may be
  - Individual consumers, groups, or the Society as a whole
  - Informed-decision-makers OR uninformed yet impacted (requiring advocacy groups/individuals on their behalf – includes other species, environment, and the unawares, underprivileged, disadvantaged people/communities)
- Experimental Impact may affect
  - Health, safety, environment
  - Welfare
    - Social status, social structure, culture, education, lifestyle, personal empowerment/situations, income and wealth distribution, *etc.*

Conscientiousness in Professionalism

- Conscientious
  - (of a person) wishing to do what is right, especially to do one's work or duty well and thoroughly
- Conscientiousness requires a determination to do what is right (for the Society, Profession, and Employer/ Organisation/ Management)
  - "Self-regulation"
  - Relevant Information is needed to be conscientious – comes through advanced expertise – It's the duty of the professional to be first informed
    - "Seek to first understand and then to be understood" (Stephen Covey's "7 Habits of Highly Effective People")
- Conscientiousness requires respect towards others' rights

Engineers as Responsible Experimenters

- An engineer is never a loner — always a team-player!
- Engineer in the Society :: Technology Enabler/ Provider/ Enhancer/ Facilitator
  - "Guardians of Public Interest"
  - Shares responsibility with (organisation) management, public, …
  - Possesses advanced expertise to monitor and execute projects, to identify and manage risks, to help clients, end-users, and other people make informed decisions, …
- Such a social responsibility requires engineers to possess
  - Conscientiousness
  - Accountability
  - Moral Autonomy

Relevant Information

- An engineer is expected to have prior information such that
  - A reasonable level of safety is assured
  - Rights of all stakeholders are respected and protected
  - Informed consent of the stakeholders is obtained as may be relevant
  - Outcomes are predicted and Risks are identified to the best knowledge
  - Net benefits are maximised
  - Team members are chosen so as to ensure all “professionalism” that is expected of oneself
  - Advice is sought from other experts as is necessary

Accountability and Answerability of Autonomous Technologies

Accountability

- Answerability + blameworthiness + liability + expectation of account-giving
- Acknowledgment and assumption of responsibility for actions, products, decisions, and policies including administration, governance, and implementation within the scope of the role or employment position and encompassing the obligation to report, explain and be answerable for resulting consequences
- Researcher accountability: being cognizant of, and take some responsibility for, the potential impact of their ways of doing research – and of writing it up – on the social fields of which the research is part.

Wikipedia, "Accountability", https://en.wikipedia.org/wiki/Accountability

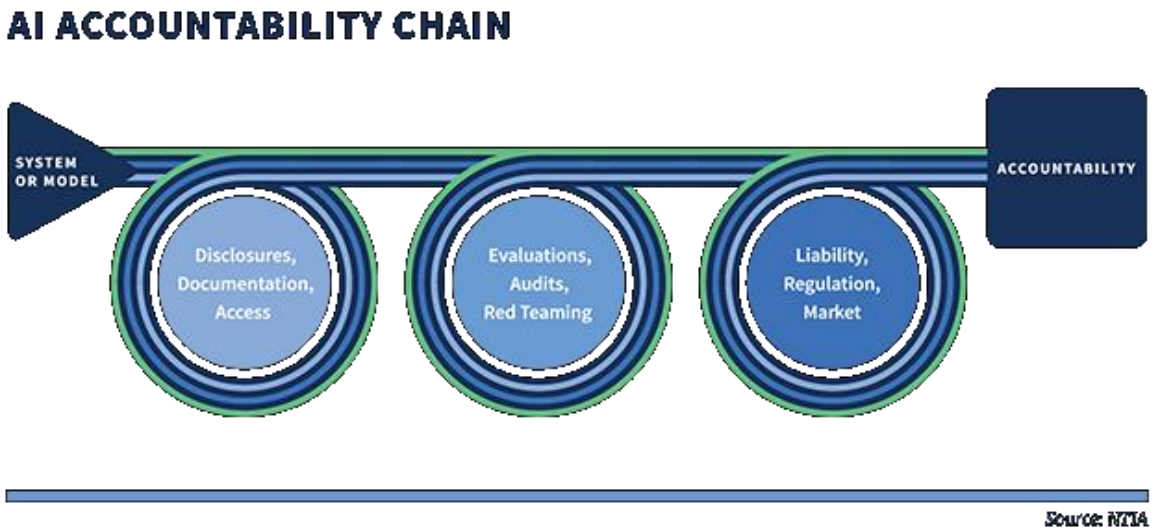
Fundamental Aspects of Trust

- Cooperation
- Reciprocity
- Sharing of information
- Reliance
- Confidence through credential exchange
- Accountability

Weber, R., "Ethics in the Internet Environment", Global Commission on Internet Governance, Paper Series No. 39, Jul. 2016 (CC-BY-3.0)

"Accountability Gap"

- Gap between technical features and social context of deployment
- Formal mechanisms of accountability to be developed and ensured: authority recognition, compliance, enforcement, reporting, …
- "Trustworthy AI" attributes: valid and reliable, safe, secure and resilient, privacy-enhanced, explainable and interpretable, accountable and transparent, and fair with harmful bias managed
- Data protection and privacy are most important



https://link.springer.com/article/10.1007/s00146-023-01635-y https://doi.org/10.1093/oxfordhb/9780197579329.013.10
https://www.ntia.gov/issues/artificial-intelligence/ai-accountability-policy-report/overview

Social Innovation

Social Innovation



Characteristics of Social Innovation

- Dynamic process of strategically developing and implementing inventive ideas, strategies, or interventions aimed at proactively addressing prevalent social issues and instigating positive, transformative change
  - Microfinance, renewable energy, digital enablement, open-source software, circular economy, …
- Helps achieve various SDG
- Can be for profit too!

https://ideascale.com/blog/what-is-social-innovation

Best Practices of Social Innovation

- Deeply Understand the Problem
- Embrace Co-Creation and Collaboration
- Empower and Include Beneficiaries
- Adopt a Systems Thinking Approach
- Build Partnerships and Networks
- Iterate and Learn from Failure
- Seek Sustainable Funding Models
- Advocate for Policy Change
- Foster a Culture of Innovation
- Share Knowledge and Scale Impact

https://ideascale.com/blog/what-is-social-innovation

Guidance through Professional Societies

Role of Professional Societies

- Bodies or organizations recognized by Government
- Members must be engineering professionals/ students/ managers

- As Caretakers:
  - Provide Codes of Ethics for professionals discharging duties
  - Update the knowledge of engineers through journals, newsletters, conferences, etc.
  - Create/ Enable/ Facilitate forums for engineers to meet and discuss common issues
  - Conduct ethics workshops and continuing education programmes for engineers
  - Resolve conflicts between engineers and employers
  - Protect rights of engineers
  - Provide social status and public recognition to engineers
- As Controlling Authorities:
  - Fight legal battles for engineers in some situations and make policy changes possible
  - Violations are seriously viewed and membership is revoked

Raghavan, M.P., "Professional Ethics and Human Values", Scitech Publications India Pvt. Ltd., 2013